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RESULTS OF A PALAEOMAGNETIC RECONNAISSANCE OF THE NEW HEBRIDES AND NEW CALEDONIA

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SUMMARY

Lavas and tuffs from islands in the New Hebrides are found to be suitable for palaeomagnetic research which suggests that many of the volcanic islands are of Pleistocene rather than Pliocene age and have not undergone detectable rotation since their formation; however, the number of samples is too few for quantitative estimates to be made of either age or rotations. Rocks from New Caledonia show variable N.R.M. characteristics, most being magnetically unstable, but it is possible that detailed palaeomagnetic analysis of a more extensive collection may yield data which will limit the hypotheses on the age and formation of ultrabasic intrusions on the island.

INTRODUCTION

Various islands in the New Hebrides and New Caledonia were visited in 1960 as a part of a palaeomagnetic reconnaissance study of Late Cenozoic lavas in the Pacific Basin (Tarling, 1962, 1963, 1965, 1966a,b; McDougall and Tarling, 1963, 1964). In both areas the main object was to determine the suitability of the rocks for palaeomagnetic work rather than to attempt a definitive study; the sampling was therefore extensive rather than intensive. Geographically and geologically the two areas are quite distinct and this is reflected in the palaeomagnetic results as the New Hebridean rocks appear particularly good for palaeomagnetic research, while the New Caledonian rocks will require extremely careful sampling and analysis if useful palaeomagnetic properties are to be isolated. In view of these differences, the two island groups are discussed separately.

NEW HEBRIDES

Introduction, geology and sampling

The Condominium of the New Hebrides (14,760 km^2) comprises about 80 islands forming a Y-shaped chain (Fig.1). The islands are generally rugged and the vegetation is dense, rendering geological mapping difficult. Most of the early geological reports are of eruptions noted by passing

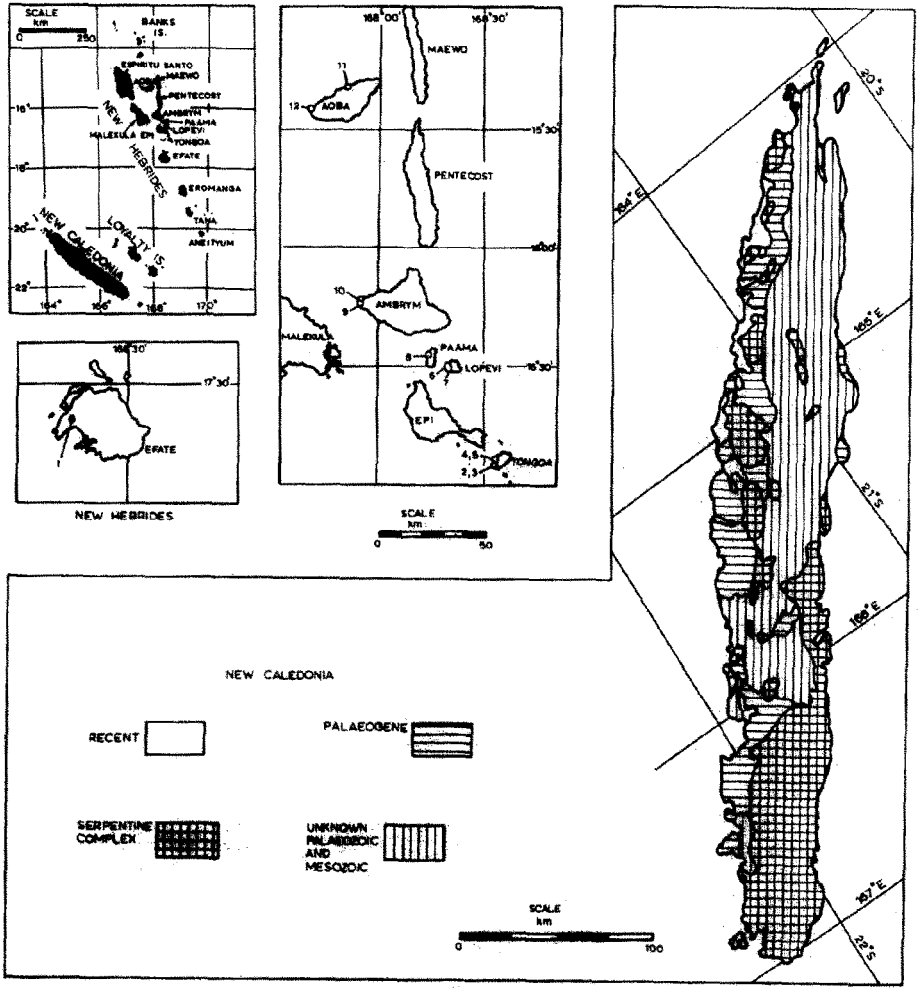


Fig.1. Sampling sites in the New Hebrides and New Caledonia.

exploration vessels and most of the geologists visiting the islands in the twentieth century have also been primarily interested in vulcanicity. Obellianne (1958) has made a general reconnaissance of Santo, Maewo, Pentecoste and Efate and Rahdon (1960) made a survey of Malekula. The Geological Survey of the New Hebrides was established in 1959 and geological mapping is currently in progress (Williams and Warden, 1964).

The outer, eastern chain appears to be formed predominantly of andesitic basalts and tuffs with some basic and ultrabasic intrusions. This contrasts with the inner, western chain which is predominantly andesitic

tuffs and lavas with associated ultrabasic intrusions, e.g., serpentinites and gabbros. The age of the islands is rather uncertain; the oldest rocks are pre-Miocene, probably Oligocene, and the major intrusions are attributed to the Upper and Middle Miocene. The majority of the islands are composed of tuffs and lavas ranging in age from about mid-Pliocene to the present; active volcanoes being on Ambrym, Lopevi, Tanna and in the Banks Islands and off the island of Epi.

Twelve sites were visited on six islands (Fig.1); 37 hand samples were collected from natural exposures, mostly along the sea coasts. These samples were drilled in the laboratory and the natural remanent magnetization (N.M.R.) of 75 specimens was measured, using astatic magnetometers (Blackett, 1952; Collinson et al., 1957), with an accuracy of 3° in direction and 5% in intensity. The exact age of the samples is not known although samples from Ambrym were thought to be from a lava and associated tuff erupted in 1913, and samples from Lopevi are almost certainly of historic age. Radioactive age determinations have subsequently been obtained (N.H. Gale, personal communication, 1966) from a sample from site 3 on Tongoa (NM7b) and site 12 on Aoba (NM35b). These yield ages of 0.26 ± 0.18 and 1.7 ± 0.65 m.y. (million years) respectively.

Stability of remanence

The stability of the N.R.M. was studied using laboratory methods. The standard field tests, outlined by Cox and Doell (1960), could not be applied to this collection.

Thermal stability

Five specimens were subjected to partial thermal demagnetization in incremental temperatures up to 650°C using the apparatus and methods outlined by Irving et al. (1961a). The directions of N.R.M. remained approximately constant between 100 and 500°C in the specimens from Tongoa, Lopevi (Fig.2), and Ambrym, but the specimen from Aoba (Fig.2) showed only poorly consistent direction in temperatures between 20 and 300°C. The intensity of N.R.M. remained higher than 70% of its initial value during the temperature range over which the direction remained consistent but decreased rapidly on approaching the Curie point (p. 61). The susceptibility varied only slightly throughout the experiment suggesting that little or no chemical change took place in the ferromagnetic minerals during the experiment.

Alternating magnetic field stability

Fourteen lava and four tuff specimens were partially demagnetized in incremental magnetic fields up to 750 Oe (peak) using the apparatus and methods described by Irving et al. (1961b). The lava specimens from Tongoa, Lopevi (Fig.2) and Ambrym showed only minor changes in direction in weak fields and then retained approximately constant directions up to the peak field. Specimens from Aoba were variable, one showing constant directions up to 450 Oe and two showing little or no consistency in direction over any range of applied field. Specimens from Paama also behaved variably, one specimen showing a constant direction up to 750 Oe and two others showing

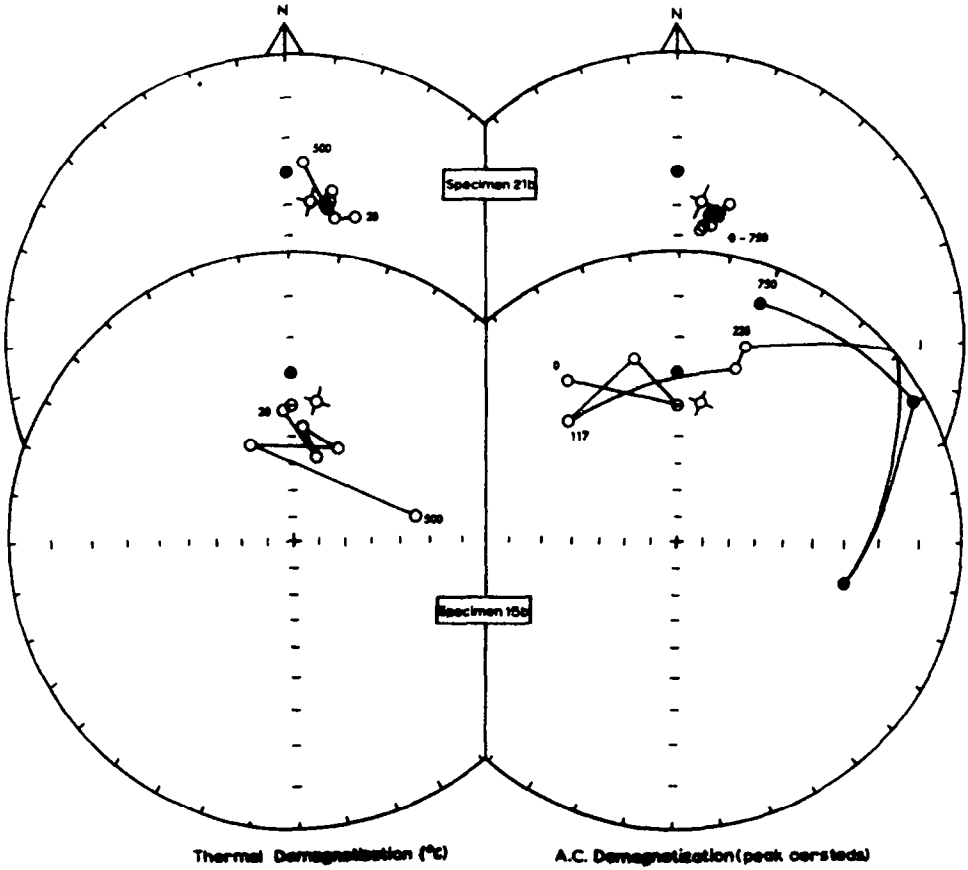


Fig.2. Thermal and magnetic stability of some New Hebridean lavas. ⊕ = axial geocentric dipole field in the New Hebrides; ⊹ = present earth's magnetic field in the New Hebrides. Examples: specimen 21b from Paama and specimen 15b from Lopevi.

considerable variation in direction in successive fields (Fig.2). Tuff specimens from Efate showed only moderate consistency up to 300 Oe and became inconsistent in higher fields. In general, all specimens exhibiting stable directions of remanence were only gradually reduced in intensity, while specimens with inconsistent directions over various ranges of applied field tended to loose most of their initial intensity in low fields.

Internal consistency

Eleven of the twelve sites showed either little change from the initial direction or had mean site directions which were significant at a 95% level of confidence after partial demagnetization in 150 Oe alternating magnetic field (Table I). The mean site directions also showed internal consistency although the sites were up to 200 km apart. However, specimens of the Paama

lava were internally inconsistent and were not in agreement with the other site directions.

Evaluation of the stability evidence

The lavas from all islands, except Paama, appear to have a N.R.M. of high stability in either thermal or magnetic fields and their directional consistency suggests that these specimens are likely to have retained their initial direction, with a small superimposed secondary component of low coercivity. However, the lavas from the site on Paama are thermally and magnetically unstable. The tuff sample from Efate shows only moderate stability although it is clear that this material is suitable for future palaeomagnetic studies in these islands.

Palaeomagnetic directions and their interpretation

In order to reduce the effect of low coercivity components, all specimens were partially demagnetized in a 150 Oe magnetic field. The specimen directions were averaged vectorally (Fisher, 1953) to obtain sample directions (Fig.3), and the sample directions were then averaged vectorally to obtain mean site directions (Table I). The mean site directions of the lavas

TABEL I

Mean New Hebridean site directions[1]

Site	No. of samples	No. of specimens	Palaeomagnetic directions						Virtual pole position
			Initial		after partial demagnetization in 150 Oe magnetic field				
			D	I	D	I	k	α	
Efate									
1*	1	4	(193	+60)	(184	+41)	-	-	(83°N 136°E)
Tongoa									
2	3	6	341	-22	342	-23	9	45	72°N 92°E
3	3	6	5	-22	0	-22	147	10	85°N 167°E
4	3	6	3	-20	9	-18	23	27	79°N 143°W
5	3	6	4	-23	4	-18	21	28	82°N 165°W
Lopevi									
6	3	6	15	-28	15	-27	52	17	76°N 108°W
7	2	4	3	-35	2	-34	812	9	87°N 53°W
Paama									
8	4	8	-	-	-	-	-	-	- -
Ambrym (1913 eruption)									
9	5	9	14	-31	13	-21	52	13	76°N 125°W
10*	1	2	(8	-35)	(350	-35)	-	-	(80°N 59°E)
Aoba									
11	4	8	314	-39	336	-33	18	22	67°N 69°E
12	5	10	349	-15	1	+1	40	12	74°N 165°E

[1]D, I = resp. mean declination and inclination calculated from sample directions; not given if not significant at a 95% level (Watson, 1956; Vincenz and Bruckshaw, 1960) and given in parentheses if only one sample per site. K, α = resp. estimate of precision and semi-angle of cone of 95% confidence (Fisher, 1953).
*Tuff samples.

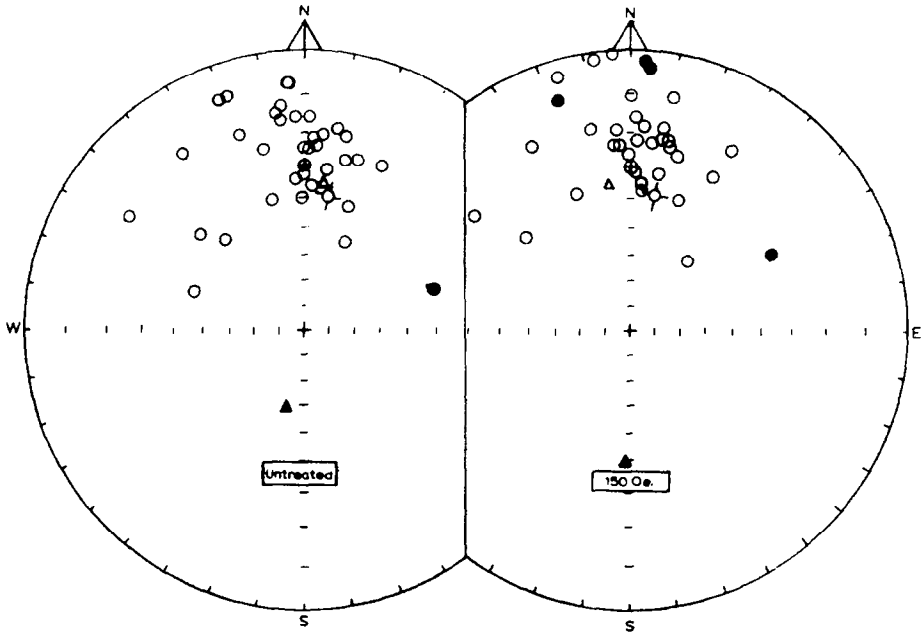


Fig.3. Sample directions of N.R.M. in the New Hebrides.
Key:
○ ● = lava } solid positive inclination.
△ ▲ = tuff } hollow negative inclination.
⊕ = axial geocentric dipole field.
◇ = present earth's magnetic field.

were all significant at a 95% level (Watson, 1956; Vincenz and Bruckshaw, 1960) with the exception of the site on Paama which is omitted from subsequent discussion on the basis of its instability in laboratory tests (p. 57). The overall mean direction of the lavas (1, - 22; α = 10°) differs significantly from the 1960 earth's magnetic field direction in the New Hebrides (11, -39), but does not differ significantly from the axial dipole field (0, -30). This suggests that although the number of sites (nine) is few, the time interval which they represent is long enough for secular variation to be averaged out and that there has been little or no tectonic rotation of the individual sites since their acquisition of N.R.M. The normal polarity of all sites would be consistent with a Middle to Late Pleistocene age but the possibility that more than one normal polarity zone has been sampled cannot be ruled out. However much more detailed sampling, combined with better age estimates, is necessary before small tectonic movements would be detected with any certainty.

The submarine tuff from Efate is reversely magnetized, having a N.R.M. direction displaced by some 160–170° from that of the mean direction of the lava sites. However, one sample cannot be used for interpretation other that the suitability of the material for future work. The tuff sample from Ambrym has, especially before demagnetization, a direction similar

to the lava of the same age; correction for the very steep subaerial bedding moves the tuff N.R.M. direction well away from that of the lava, thereby suggesting that the remanence is of low stability but acquired while the tuff was a hot deposit and is therefore of thermoremanent origin.

Other magnetic properties

Intensity and susceptibility

The arithmetic mean intensity of the lavas is $7.54 \cdot 10^{-3}$ Gauss and the geometric mean is $5.19 \cdot 10^{-3}$ Gauss. This latter, corrected to its palaeo-equatorial value, assuming the dipole formula, is $4.92 \cdot 10^{-3}$ Gauss with a standard deviation limit from 11.17 and $2.17 \cdot 10^{-3}$ Gauss. The intensity is therefore slightly higher than the average intensity of most Cenozoic oceanic basalts and higher than average intensities reported in continental Cenozoic basalts (Tarling, 1966b). The arithmetic average susceptibility of the lavas is $4.59 \cdot 10^{-3}$ Gauss/Oe and the geometric mean is $4.20 \cdot 10^{-3}$ with standard deviation limit from 6.68 to $2.64 \cdot 10^{-3}$ Gauss/Oe. This value is slightly higher than the susceptibilities of most Cenozoic basalts (Tarling, 1966b), and the range of observed values is more restricted.

The intensity and susceptibility of the Efate subaerial tuff are $0.05 \cdot 10^{-3}$ Gauss and $0.04 \cdot 10^{-3}$ Gauss/Oe, and the Ambrym tuff values are $0.17 \cdot 10^{-3}$ Gauss and $7.57 \cdot 10^{-3}$ Gauss/Oe.

Thermoremanent magnetization (T.R.M.)

Two specimens, from Lopevi and Ambrym, were given a T.R.M. by heating to 675°C and allowing them to cool in the earth's magnetic field (0.59 Oe). They both acquired remanent directions which are the same, within experimental error, as the applied field suggesting that there is little or no anisotropy of T.R.M. present in these rocks. The intensity of T.R.M. was higher than that of N.R.M., the ratio of N.R.M. to T.R.M. being 0.59 in the Lopevi specimen and 0.79 in the Lopevi specimen; if the T.R.M. is corrected to that which would be acquired in 0.45 Oe, the present total intensity of the earth's magnetic field in the New Hebrides, then these ratios are 0.76 and 1.02. These ratios suggest that the earth's magnetic field may have been very slightly greater than its present value in Ambrym in 1913. The Lopevi specimen may have cooled in a field weaker than 0.45 Oe, but it is more probable that the original intensity has decayed since its acquisition and therefore the present N.R.M. does not represent the original intensity of T.R.M. However, no reliable interpretation can be made on such few data.

Curie points

Curie points were determined to ± 15°C, using a thermomagnetic balance (Chevallier and Pierre, 1932), for all specimens which were thermally demagnetized. All showed reversible heating and cooling curves with Curie points between 500 and 525°C which is consistent with a titanomagnetite composition of approximately 90% (Fe_3O_4), 10% ($TiFe_2O_4$). The Curie points of the Efate tuff were all determined as 475°C suggesting that the composition of titanomagnetites in the tuff is slightly different to those in the lavas.

Conclusions

The andesitic basalts of the New Hebrides appear to be very stable and therefore suitable for palaeomagnetic research. The present study suggests that the lavas sampled are of Pleistocene age with little or no tectonic rotation between the different sites. The radioactive age determinations are consistent with the polarity zonation so far established for the Plio–Pleistocene as both ages correspond to periods of normal polarity established elsewhere (Cox et al., 1964). However, the Tongoa specimen is thought to be younger than the calculated age of of 1.7 m.y. and probably belongs to the present polarity period. However, the survey is much too restricted to allow conclusions to be drawn although the detailed tectonic behaviour of this active part of the earth's crust is of particular importance in orogenic theories.

NEW CALEDONIA

Introduction and geology

New Caledonia is one of the largest islands in the Pacific, covering 16.100 km^2 (Fig.1). It is in the form of a long ellipsoid, 50 by 400 km., and it is mountainous, rising to 1.600 m in many places. The geology is complex, ranging from Palaeozoic metamorphic rocks through to present day reef formation and including at least two strong tectonic phases. Considerable geological and economic interest is centred on the Serpentine Complex, attributed to the Oligocene. The Complex consists mainly of harzburgite (about 90%), dunites and gabbros, exhibiting varying degrees of serpentinization and containing valuable nickel and chrome ores. However, there is not general agreement on the form or exact age of the Complex, or of its constituent parts (Chételat, 1947; Avias 1949; Maxwell, 1949; Routhier 1953, Caillère et al. 1956) and it was hoped that palaeomagnetic techniques might help in determining the age and structural relationship between the dateable lavas of Palaeogene age.

In order to test the possibility of applying these techniques hand samples were obtained from 35 sites (Fig.1) in (a) the Serpentine Complex, (b) the Palaeogene lavas, (c) the Jurassic–Cretaceous "formation à charbon" and (d) the Palaeozoic metamorphic "phyllades". Specimens were cut from these samples and measured in the same way as the New Hebridean lavas (p. 57).

Stability of N.R.M.

Serpentine Complex

Eleven specimens were subjected to partial demagnetization in a.c. magnetic fields up to 750 Oe (p. 57). Six specimens showed random directions in all applied fields, one specimen showed constant directions (Fig.4) between 37 and 225 Oe; other specimens showed only poor consistency over even shorter ranges of applied field. Four specimens were subjected to thermal demagnetization (p. 57) but only poor consistency in direction was

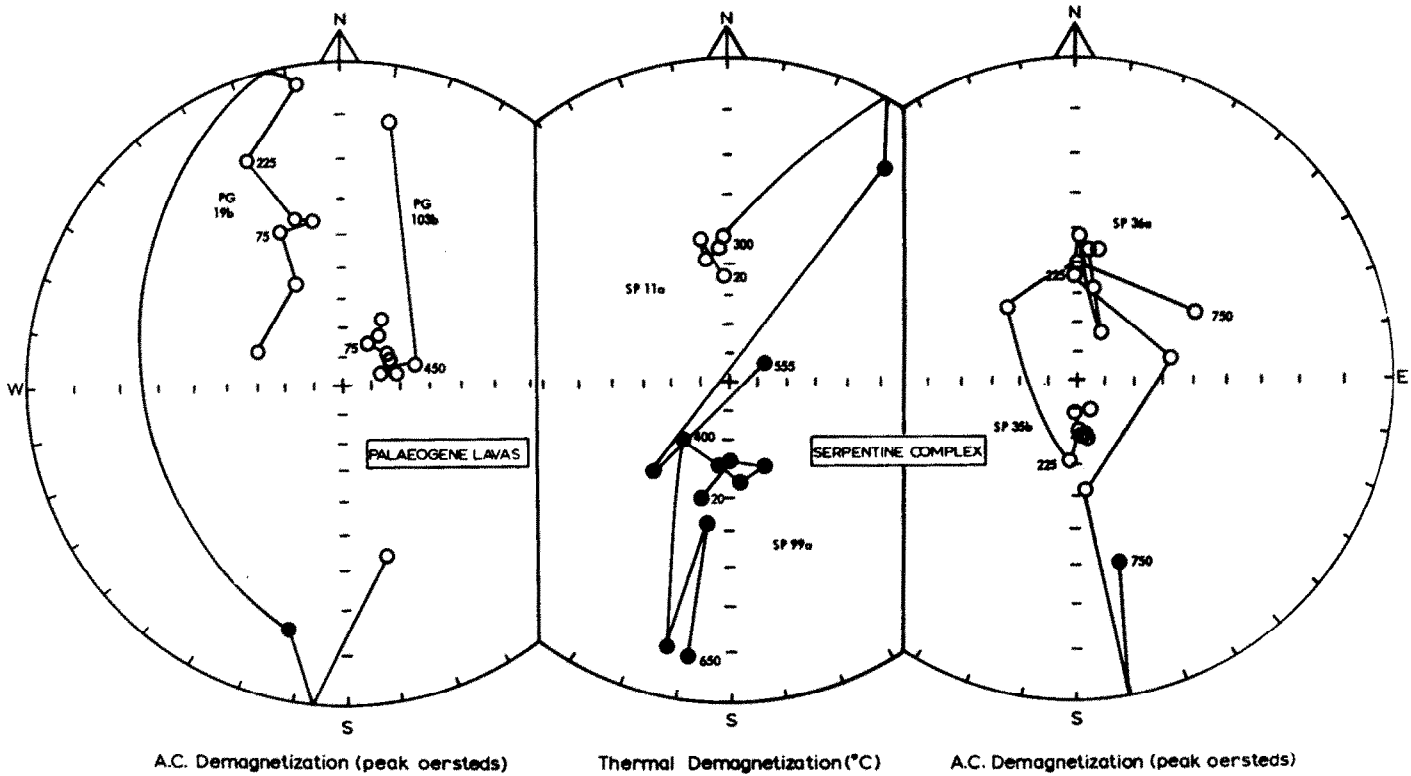


Fig.4. Magnetic stability of some New Caledonian specimens.

observed between 100 and 400°C (Fig.4). It is considered that these results show that most of the Serpentine Complex is of low stability although the more basic specimens appear to be more stable than those which are highly serpentinized.

Palaeogene lavas

Fourteen specimens were partially a.c. demagnetized; four showed N.R.M. directions which were random in successive fields; five showed very consistent directions (Fig.4) up to 200 or 450 Oe and other specimens showing only moderate consistency over smaller ranges of applied fields (Fig.4). The Palaeogene lavas are therefore considered to be of varying stability.

Metamorphic "phyllades" and "formation à charbon"

Two specimens from each formation were partially a.c. demagnetized but their intensity of N.R.M. was very low before demagnetization and became unmeasurable in only moderate applied fields.

Palaeomagnetic directions

All specimens were partially demagnetized in a peak field of 75 Oe in order to reduce the effects of low coercivity components of N.R.M. In some cases a higher field may have been more effective in isolating the primary N.R.M. component but stability tests (p. 62) suggested many specimens became randomly magnetized in only moderate a.c. fields. Only directions of N.R.M. will be considered after partial demagnetization and these have been combined vectorally and their significance tested as for the New Hebridean lavas (p. 59).

Specimens from the metamorphic "phyllades" and the "formation à charbon" were only weakly magnetic and showed widely scattered remanence

TABLE II

Significant (95%) site directions in New Caledonia after partial demagnetization[1]

Site	No. of samples	No. of specimens	Palaeomagnetic directions initial D	I	after partial demagnetization in 150 0e magnetic field D	I	k	α	Virtual pole position
Serpentine Complex									
4	3	5	299	-56	269	-65	11	39	15°N 28°E
5	5	9	0	-74	18	-84	6	35	33°N 20°W
10	3	6	181	+18	176	+23	16	31	80°S 37°W
Palaeogene lavas									
1	2	4	-	-	0	-46	101	25	84°N 10°W
4	3	3	30	-37	46	-31	22	27	46°N 102°W
9	7	10	334	-27	337	-25	4	30	49°N 130°E
11	3	5	308	+28	310	+26	18	30	30°N 107°E
13	3	5	313	+17	311	+13	24	26	34°N 102°E

[1]D, I = resp. mean declination and inclination calculated from sample directions; not given if not significant at a 95% level (Watson, 1956; Vincenz and Bruckshaw, 1960). k, α = resp. estimate of precision and semi-angle of cone of 95% confidence (Fisher, 1953).

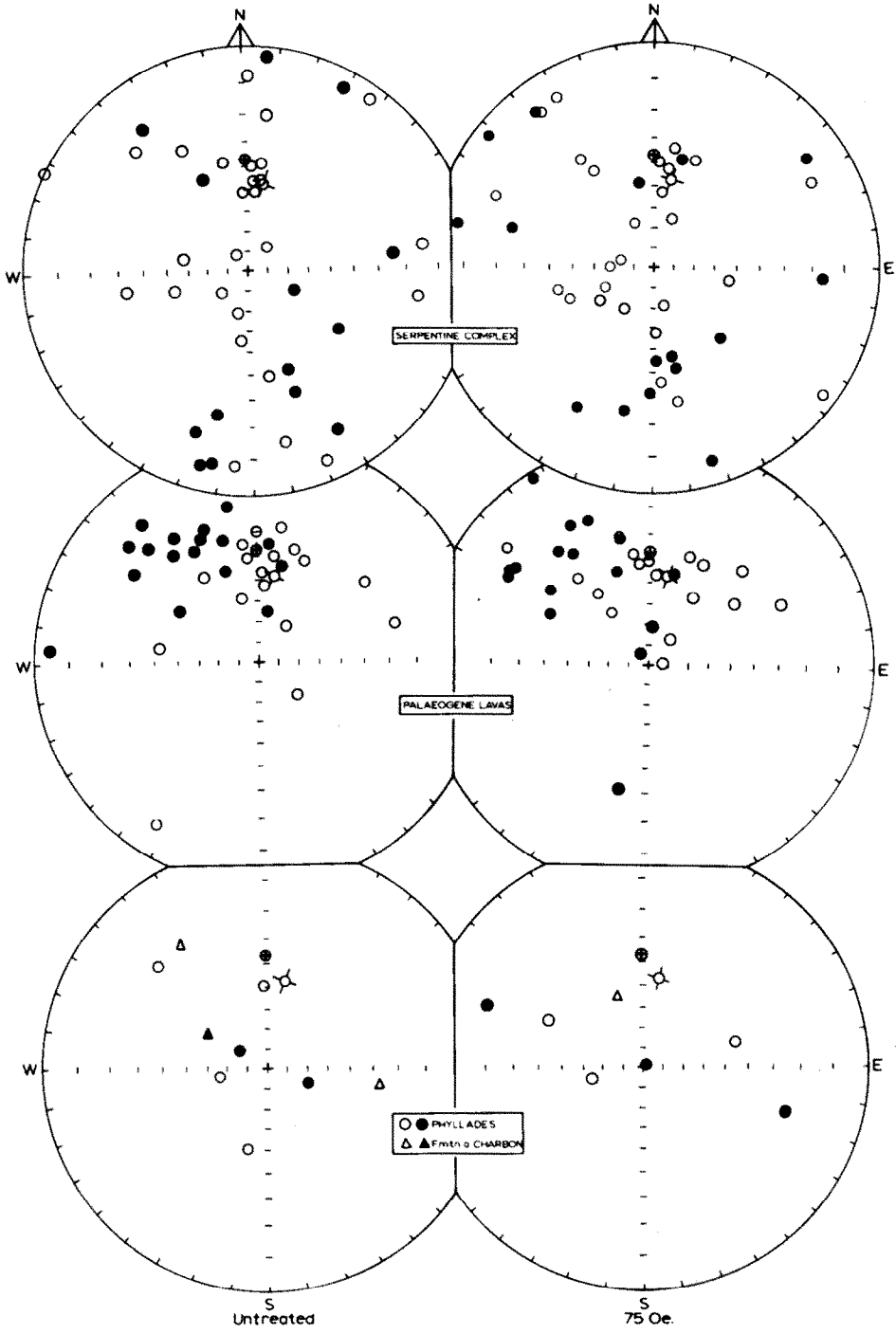


Fig.5. Sample directions of N.R.M. in New Caledonia. Solid symbols = positive inclination; open symbols = negative inclination.

directions (Fig.5). The directions of N.R.M. in samples from the Serpentine Complex are also widely scattered (Fig.5) and only three of the fourteen sites have significant (95%) directions. These significant site directions (Table II) show considerable variation which may be attributable to either instability or tectonic movements. The directions of N.R.M. of Palaeogene lava samples show wide scatter (Fig.5) and only five of the thirteen sites have significant directions (Table II). These sites all have northerly declinations but still show considerable variation which may be attributed as in the Serpentine Complex, to either instability or tectonic movements but the data are insufficient for further analysis and there is no overall significant direction.

Other magnetic properties

Intensity and susceptibility

The arithmetic and geometric (Tarling, 1966b) averages and standard deviations of all specimens from each formation are given in Table III. The metamorphic "phyllades" and the "formation à charbon" appear to have very weak intensity and susceptibility, although the number of samples is small. The Palaeogene lavas and the Serpentine Complex have similar low susceptibilities although the N.R.M. intensity in the Serpentine Complex is about three times higher and more broadly distributed than in the Palaeogene lavas.

Thermoremanent magnetization

Four specimens were given a T.R.M. (p. 61) in the earth's magnetic field (0.59 Oe). One specimen from the Serpentine Complex and one from the Palaeogene lavas showed T.R.M. directions which differed significantly from that of the applied field. The other two, also from the Serpentine Complex and Palaeogene lavas, acquired directions similar to the applied field. The intensity of T.R.M. was much higher than the N.R.M. both at room

TABLE III

Intensity and susceptibility of New Caledonian specimens

Formations	No. of specimens	Intensity (10^{-3} Gauss) A.M.[1]	G.M.[2]	G.M. + stand.dev.	G.M. - stand.dev.
Serpentine Complex	75	805	204	1,752	24
Palaeogene lavas	51	434	65	664	6
Formation à charbon	8	2	0.1	3	0.01
Metamorphic "phyllades"	14	3	1	6	0.3
		Susceptibility (10^{-3} Gauss)			
		A.M.	G.M.	G.M. + stand.dev.	G.M. - stand.dev.
Serpentine Complex	44	500	150	1,120	20
Palaeogene lavas	35	1,460	130	1,910	10
Formation à charbon	8	320	30	230	3
Metamorphic "phyllades"	6	10	5	30	1

[1]A.M. = arithmetic mean.
[2]G.M. = geometric mean.

temperature and over the same higher temperature ranges, but this is thought to reflect a decay of N.R.M. rather than a smaller ambient field when the rocks acquired their magnetization.

Curie points

The Curie points of seven Serpentine Complex specimens were determined (p. 61) and all fell within the region of 500–525°C on heating, but usually showed lower Curie points on cooling, i.e., the thermomagnetic curves are irreversible. Five specimens of Palaeogene lava had Curie points between 475 and 525°C, and similar Curie points were observable on cooling although the full curve was not reversible.

Conclusions

The reconnaissance survey suggests that neither the "phyllades" nor "formation à charbon" are likely to yield useable palaeomagnetic results. There is considerable instability present in both the Serpentine Complex and the Palaeogene lavas. It is probable that a detailed survey, combined with stringent a.c. or thermal demagnetization and tests for anisotropy, could reveal the original direction of the earth's magnetic field at the time that the lavas and serpentinites acquired their remanence, and such directions could possibly be used for dating within the Serpentine Complex.

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